



GUIDELINE

Analgesia and Sedation in Paediatric Critical Care

Scope (Staff):	Paediatric Critical Care, Clinical Staff
Scope (Area):	Paediatric Critical Care, PCH

Child Safe Organisation Statement of Commitment

CAHS commits to being a child safe organisation by applying the National Principles for Child Safe Organisations. This is a commitment to a strong culture supported by robust policies and procedures to reduce the likelihood of harm to children and young people.

This document should be read in conjunction with this [disclaimer](#)

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Aim

1. To standardise the assessment and management of pain and sedation level in patients admitted to the Paediatric Critical Care (PCC) unit using validated tools.
2. To reduce the incidence of complications related to pain, anxiety management and sedation such as delirium, withdrawal, and increased duration of mechanical ventilation.

Background

Sedation and analgesia are an important part of the management of critically ill children. Although necessary these medications are associated with many adverse effects such as disruption of circadian rhythm, altered sleep, delirium, potential neurotoxicity and immunosuppression.¹ Use of sedation scores, identification of individualised patient sedation targets and titration of sedative agents to achieve these targets allows for individualised approach to sedation and has been recommended in international consensus guidelines to minimise the use of these medications.² Use of protocolised guidelines have been demonstrated to reduce patient exposure to sedatives, intensive care unit (ICU) length of stay, duration of sedation and prevalence of drug withdrawal.^{3,4}

Risk

Failure to adhere to these guidelines can result in undersedation, oversedation, inadequate or over treatment of pain. Undersedation may result in patient-ventilator dyssynchrony, anxiety, post-traumatic stress disorder (PTSD) and unplanned extubation or removal of lines.² Oversedation can contribute to short term complications such as cardiorespiratory instability, gastrointestinal disturbance and depression of the immune system as well as longer term complications such as development of withdrawal, delirium, increased PCC and hospital length of stay, and long term neurocognitive effects.^{1,5,6} Uncontrolled pain can lead to anxiety and distress for the patient and their family, delayed wound healing, hyperglycaemia, the development of chronic pain or PTSD. Overtreatment of pain can result in development of hyperalgesia, oversedation and risk of dependence and withdrawal.⁷

Definitions

Analgesia: Blunting or eradicating the sensation of pain or other noxious stimuli.

Context sensitive half time: the time required for blood or plasma concentrations of a drug to decrease by 50% after discontinuation of the infusion

Delirium: Syndrome of acute brain dysfunction common in critically unwell children characterised by fluctuating symptoms, disorientation and hyperactivity or hypoactivity.

Pain: An unpleasant sensory and emotional experience associated with actual or potential tissue damage.

Sedation: A drug induced decrease in the level of consciousness. This ranges from anxiolysis through to general anaesthesia where the patient has complete loss of consciousness.

Tolerance: Decrease in drug effect over time or need to increase dose to achieve the same therapeutic effects. This occurs due to down-regulation of receptors.

Withdrawal: Physical signs and symptoms that occur in response to cessation or rapid tapering of a drug. In the case of withdrawal from opioids and/or benzodiazepines this may present with symptoms of central nervous system irritability, gastrointestinal dysfunction and autonomic dysfunction. Generally managed with slow wean from medications.

Note: There is significant overlap between pain, oversedation, undersedation, withdrawal and delirium (see figure 1). Use of specific assessment tools can help determine the cause of the patient's distress.

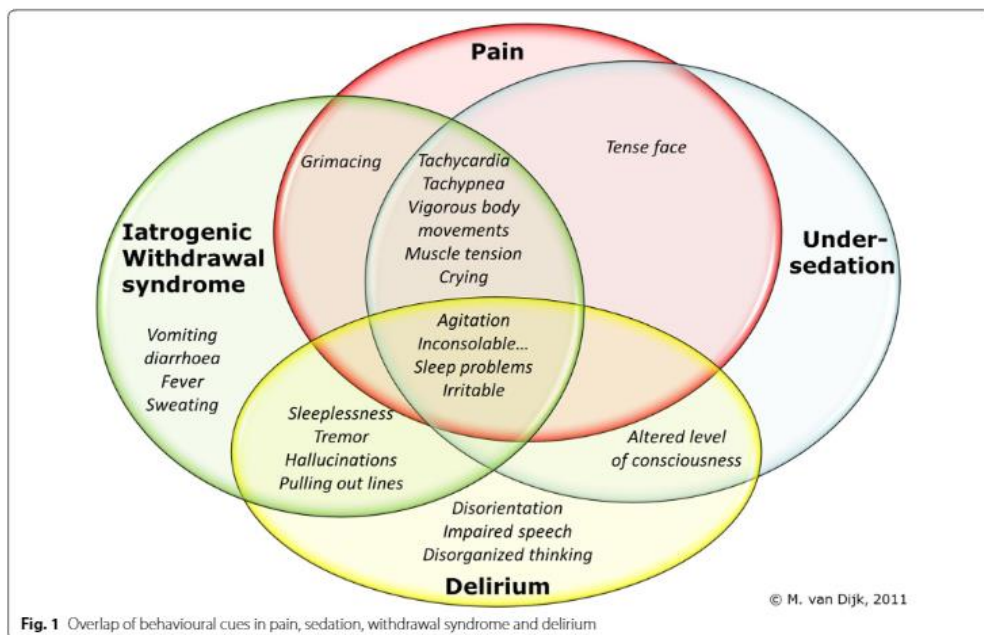


Fig. 1 Overlap of behavioural cues in pain, sedation, withdrawal syndrome and delirium

Figure 1 – overlap of behavioural cues in pain, sedation, withdrawal syndrome and delirium²

Key Points

- Always optimise non-pharmacologic management of pain, distress and anxiety in patients requiring analgesia or sedation. Pain and agitation can be exacerbated by emotional distress and anxiety. Nonpharmacological measures will reduce the requirement for pain and sedation medications. Refer to Keeping Kids in No Distress (KKIND) team where appropriate.
- Delirium is associated with increased morbidity and mortality. Maintaining day-night cycle through management of noise and light as well as clustering of cares helps to minimise the development of sleep disturbances and delirium.⁸
- Pain should be controlled prior to increasing sedative drugs. Adequate use of analgesia will generally reduce the sedative requirement and reduce the risk of oversedation and the associated complications of oversedation.⁹
- Sedative medications should be titrated to a target State Behavioural Scale (SBS) score. This should be set by the consultant on the ward round each morning. In the majority of ventilated patients, light sedation with an SBS score of -1 to 0 will be targeted to achieve a calm and co-operative patient. Deep sedation will be required in some patients such as those with severe pulmonary or intracranial hypertension or those requiring neuromuscular blockade.
- The minimum dose of sedative required to achieve the target score should be used.

- At least 3 bolus doses should be administered prior to an increase in rate of a continuous infusion. This is because it takes around 5 half-lives for a drug to achieve a steady state. Administration of a loading dose or repeated bolus doses will reduce time taken to reach steady state thus reducing risk of oversedation from repeated increases in rate prior to achievement of steady state.
- The approach to pain and sedation management in intubated patients is summarised in flow charts (see Appendix 1-3).
- Neuromuscular blocking agents (NMBA) may be required in critically ill patients to manage their haemodynamics, ventilation or intracranial pressure. These provide no sedation and should only be used in patients who are deeply sedated (SBS of -2 or -3 prior to use).
- Prolonged use of NMBA is associated with critical illness polyneuropathy and myopathy, particularly if used in combination with steroids or in patients with sepsis.¹⁰ To reduce this risk the minimum dose of NMBA required should be used.

Assessment Tools

Pain Assessment

Pain is assessed depending on patient's conscious level and age.

1. Intubated patients are assessed using the Multidimensional Assessment of Pain Scale (MAPS). This should be performed a minimum of 4 hourly and documented in the bedside observation chart. See [Appendix 4: MAPS](#)
2. Non-intubated children up to 8 years are assessed with the Faces, Legs, Activity, Consolability, Crying (rFLACC) score. See [Appendix 5: FLACC-R](#)
3. Non-intubated children >4 years: self-reporting. Children can self-report via the Faces Pain Scale (suitable from aged 4 years) or the Numeric pain rating scale (suitable from age 8 years). [Appendix 6: Pain Self-Report Assessment Tools.](#)

Sedation Assessment

Sedation is assessed using the SBS score in all intubated patients. Sedative medications should be titrated to a target SBS score. See [Appendix 7: State Behavioural Scale \(SBS\).](#)

Pain and Sedation Assessment in Paralysed Patients

- There are no validated tools for the assessment of pain and sedation in paralysed patients. An increase in vital signs (heart rate, blood pressure) with routine cares of >20% suggests inadequate analgesia or sedation.⁴
- Dilated pupils can be another sign that a patient is under-sedated as paralytic agents do not affect pupillary responses.

Management of Pain and Distress

Management strategies for pain and sedation are comprised of non-pharmacological and pharmacological management. Non-pharmacological management should be optimised in all patients to reduce the amount of sedation and analgesia required. Pharmacological management is focused on management of pain and sedation.

Non-pharmacological Management

Separation from parents, invasive procedures, disruption of usual day-night cycle, presence of unfamiliar people and machines and physical discomfort can all contribute to agitation. Non-pharmacological management strategies can address some of these and reduce the requirement for sedative and analgesic medication.

Non-pharmacological management strategies include:

- Manage other causes of distress discomfort e.g. hunger, urinary retention, extravasation of lines, pressure injury, gastrointestinal (GI) spasms.
- Encourage day-night routine through reduction of light and noise and clustering of cares overnight. Consider use of ear muffs.
- Encourage parental presence and familiar objects and photos in the bedspace.
- Use non-verbal communication to reassure the patient.
- Frequent repositioning to reduce patient discomfort and avoid pressure injuries
- Utilise KKKIND principles in awake children including distraction techniques for children requiring procedures.
- Consider swaddling and dummy for infants. Skin to skin contact for neonates should also be considered where clinically appropriate.
- Consider use of hot / cold pack or flannel for patients for management of pain or temperature control.

Pain Management

- Pain should be managed with a multimodal approach with optimum use of simple analgesics and non-pharmacological measures in all patients.
- Simple analgesic agents include sucrose (for infants), paracetamol and non-steroidal anti-inflammatory drugs (NSAIDs).
- All post-operative patients should have regular paracetamol charted unless contraindicated and the use of NSAIDs should be clarified with the surgical team.

Sedation Management

- In general, the lowest possible dose of sedative to keep patients safe should be used. Oversedation is more common than undersedation in paediatrics.¹¹

- With the exception of dexmedetomidine, which is known to induce a natural, sleep-like state, the sedatives traditionally used in the PICU are known to decrease restorative sleep and may contribute to the development of delirium.⁷

Specific Analgesic and Sedative Therapies used in PCC

Non-opioid analgesia

Sucrose 24%

Useful in infants for procedural pain relief (mild-moderate pain only e.g. venepuncture, heel prick, nasogastric tube (NGT) insertion). Thought to stimulate the release of endorphins that act at opioid receptors.¹¹ Used for children up to 18 months of age.

Dose:

- Preterm infant (corrected gestational age 32-40 weeks): 0.2-0.5mL per procedure (max 2.5mL/24 hours).¹²
- 0-1 month: 0.5-1mL per procedure (maximum 5mL in 24 hours).¹²
- 1-18 months: 1-2mL per procedure (<3 months: maximum 5mL in 24 hours or >3 months: maximum 10mL in 24 hours).¹²

Pharmacokinetics

- Onset: 10 seconds
- Peak effect: 2 minutes
- Additive benefit observed when combined with the use of a dummy in neonates undergoing painful procedures.¹³

Paracetamol

See [Paracetamol medication monograph](#) for further details

Paracetamol has analgesic and antipyretic properties and demonstrated efficacy as an opioid sparing agent. Unlike NSAIDs, it does not have significant anti-inflammatory activity or antiplatelet effect.¹³ It can be given IV or enterally. It should be prescribed regularly in all post-operative patients in PCC while they are receiving opioid analgesia.

Mechanism of action:

- Analgesic effect is via central inhibition of prostaglandin production via the cyclooxygenase pathway.

Pharmacokinetics:

- Peak concentrations after 45-60 minutes post oral administration and at the end of the infusion when administered IV.¹⁴
- Half-life 3-4 hours (7 hours in neonates).^{13,15}
- Extensive hepatic metabolism including to a hepatotoxic metabolite that is further metabolised by liver to nontoxic metabolites. Metabolites excreted in urine.¹⁵

- IV paracetamol may cause hypotension.¹⁶

Special Considerations

- IV administration may be preferable in the immediate post-operative period due to more rapid onset of action as well as patients unable to tolerate enteral paracetamol but be cautious in patients who are haemodynamically unstable.¹⁶
- Consider dose reduction in patients with hepatic or renal impairment
- Check LFTs after 72 hours of regular dosing (if IV) and after 1 week (if enteral).
- Risk of hepatotoxicity increased with malnutrition, prolonged fasting, fever, dehydration, young age and hepatic impairment.¹⁵
- Dose as per ideal body weight in children who are obese or overweight. Refer to [PCH Guidelines for Drug Dosing in Overweight and Obese Children 2 to 18 years of Age](#).

Non-Steroidal Anti-inflammatory Drugs (NSAIDs)

NSAIDs have analgesic, antipyretic and anti-inflammatory properties with demonstrated opioid sparing effects.¹⁶ They are more effective in the prevention of pain than the treatment of established pain.¹⁶ Ibuprofen is the main NSAID used in PCC while parecoxib is often given to patients peri-operatively as a single dose. Celecoxib is sometimes used in post-operative adolescent patients.

Mechanism of action

- The major analgesic effect is through the inhibition of prostaglandin synthesis via the cyclo-oxygenase (COX) pathway blocking the production of prostaglandins that stimulate nociceptors in the peripheral nervous system.¹⁷
- Inhibition of COX-1 results in impaired gastric cytoprotection and antiplatelet effects whereas inhibition of CoX-2 results in anti-inflammatory and analgesic action.¹⁸
- Reduction in glomerular filtration rate and renal blood flow occurs with both COX-1 and COX-2 inhibition.¹⁸
- Ibuprofen is a non-selective, balanced inhibitor of both COX-1 and COX-2 enzyme systems while celecoxib is a selective COX-2 inhibitor with no COX-1 inhibition at therapeutic concentrations.^{19,20}

Pharmacokinetics:

- Ibuprofen
 - Time to peak effect 45 minutes in fasted children and 1-2 hours in postprandial dosing.¹⁹
 - Half-life 2 hours.¹⁹
 - Extensive hepatic metabolism. Renal clearance of metabolites¹⁹

- Celecoxib
 - Time to peak effect of 3 hours.²⁰
 - Elimination half-life of 6 hours.²⁰
 - Hepatic metabolism in inactive metabolites. Excreted in faeces and urine.²⁰

Dosing

- Ibuprofen
 - ≥ 3 months: 5-10mg/kg (max dose 400mg) 8 hourly enterally.²¹
- Celecoxib
 - ≥ 2 years: 10-25 kg: 50mg bd enterally; >25 kg 100mg bd enterally.²¹²

Adverse effects:

- Gastrointestinal: Non-selective NSAIDs can cause gastric mucosal damage that may result in ulceration and bleeding. This is rare in children and ibuprofen is one of the NSAIDs associated with lowest risk of serious gastrointestinal adverse events.^{17,19,23} Risk of gastritis and gastric ulcer risk is increased with history of peptic ulcer or GI bleeding, H. pylori, inflammatory bowel disease or use of concurrent anti-coagulants.^{17,19}
- Renal toxicity: increased risk in patients who are hypovolaemic, hypotensive or have pre-existing renal impairment.¹⁹
- Bronchospasm. Risk of bronchospasm in patients with aspirin-induced asthma.¹⁹
- Bleeding risk with ibuprofen due to effect on platelet aggregation.¹⁷ Reduced risk with selective COX-2 inhibitors such as celecoxib.¹⁸
- Bone healing: conflicting evidence regarding effect on bone healing.¹⁷ It is often avoided in patients in who bone healing may be a significant issue (e.g. post spinal fusion).¹³

Special Considerations:

- Wait 12 hours post parecoxib dose before first dose of ibuprofen (or other NSAID).²⁴
- Avoid use in patients with aspirin sensitive asthma, renal impairment, gastrointestinal disease (particularly gastritis or chronic inflammatory bowel disease), dehydrated patients, patients with liver dysfunction and patients post kidney transplant.^{17,19,23} Avoid ibuprofen in patients with bleeding disorders.¹⁷
- Consult treating team prior to using in oncology, neurosurgical or orthopaedic patients.
- Commence in cardiac surgical patients the day after surgery if no bleeding concerns.

- Consider the use of a gastroprotection with a proton pump inhibitor (PPI) in patients at risk for GI adverse effects (particularly concomitant use of steroids, aspirin or history of GI bleeding) or patients on long term NSAIDs.²³
- Preferable to give with food or milk but likely safe to give to fasted children as no local irritation effect has been shown to contribute to risk of gastritis / gastric ulcer.¹⁹

Gabapentin

Structural analogue of GABA which binds to voltage gated calcium channels that control the release of excitatory neurotransmitters responsible for nociception.²⁵

Uses:

- Analgesia adjunct for postoperative or neuropathic pain.

Pharmacokinetics:

- Bioavailability: decreases as dose increases. Not affected by food.²⁶
- Time to peak effect: 2-4 hours.²⁶
- Elimination half-life 5-7 hours.²⁶
- Eliminated unchanged in urine and clearance proportional to creatinine clearance.²⁶

Dosing:

- <12 years enterally: ²⁷
 - initially 5mg/kg/dose at night
 - increasing to twice daily dosing on day 2 and three times daily dosing on day 3
 - then titrate to response to maximum 40mg/kg/day in three divided doses
- 12-18 years enterally: ²⁷
 - Initially 100-300mg nocte
 - Increase dose daily titrated to response.
 - Maximum dose 3.6g daily in three divided doses

Adverse Effects

- Gastrointestinal: nausea, vomiting, constipation²⁶
- Central nervous system (CNS): dose dependent CNS depression.²⁶ May worsen absence or myoclonic seizures.⁷ May cause dizziness, ataxia, confusion, emotional lability, depression, memory impairment.²⁶
- Respiratory: May cause respiratory depression when given with other CNS depressants (e.g. opioids, benzodiazepines).²⁶

Special Considerations:

- Consider dose reduction in renal impairment.²⁷
- Limited evidence of efficacy in post-operative or neuropathic pain in children.²⁵

Opioids

Opioids bind to opiate receptors in the central nervous system and peripheral tissues to produce potent analgesic effects with some sedation.⁷

Adverse effects

Adverse effects common to all opioids include:

- Gastrointestinal symptoms (constipation, ileus, feed intolerance).²⁸ All patients commenced on opioids should be charted for regular aperients.
- Nausea and vomiting. Manage with anti-emetic as first line (e.g. ondansetron) then consider addition of a motility agent (e.g. metoclopramide).²⁸
- Pruritus. Manage with PRN loratadine. If continues to be problematic can consider rotation to fentanyl (negligible histamine release) or consider use of low dose naloxone infusion (i.e. 0.25-2microg/kg/hr titrated to effect).^{7,13, 28,29}
- Respiratory depression and apnoea.²⁸ This is a dose dependent effect and more common in children aged < 6months due to decreased renal elimination and hepatic immaturity. Use lower starting doses in this age group.²⁹ IV naloxone 0.01-0.03mg/kg every 2-3 minutes PRN may be required for respiratory depression in non-intubated children.¹³
- Urinary retention: patients on continuous infusions of opioids should have an indwelling urinary catheter placed.¹
- Hyperalgesia, tolerance, dependence and withdrawal with prolonged use: Lower initial doses of opioid may be associated with higher rates of tolerance due to ongoing pain, hyperalgesia and windup.³⁰

Morphine (intravenous)

See [Morphine medication monograph](#) for further details

Morphine is the first line analgesic agent in most PCC patients. It has improved sedative effects compared with fentanyl and slower onset of tolerance but causes histamine release, is associated with higher incidence of haemodynamically instability and has a slower onset of action.⁷

Pharmacokinetics:

- Single bolus dose:
 - Onset of action: 5-10 minutes¹
 - Time to peak effect of 20 minutes^{4,14}
 - Duration of effect 2-4 hours^{4,14}

- Continuous infusion
 - Terminal elimination half-life of 2-3 hours in healthy children⁷
 - Prolonged half-life in children with cardiovascular instability, hepatic and renal impairment^{1,14}
- Metabolised by the liver to active and inactive metabolites that are then excreted by the kidneys.¹⁴
- Neonates and infants have reduced morphine clearance but by age of 6 months this has reached 80% of adult levels.³⁰
- Reduced efficacy in preterm neonates due to hepatic immaturity.²⁹

Dosing for intubated patients:

- Loading dose:
 - <6 months: 50microg/kg IV
 - >6 months: 50-100microg/kg IV (max 2.5-5mg) ⁴
- Continuous infusion dose: 0-60microg/kg/hr. Higher doses may be required, e.g. if they are opioid tolerant patients
- Bolus dose for pain: 20microg/kg IV (max 1mg)

Specific side effects:

- Causes histamine release which can result in hypotension and bronchoconstriction.¹² Fentanyl may be preferred agent in patients with hypotension or severe reactive airways disease.
- Hypotension also mediated via negative chronotropic and inotropic effects and a decrease in baroreceptor reflex response.¹³
- Risk for accumulation in patients with renal failure, hepatic impairment or decreased hepatic blood flow - in patients with severe low cardiac output state (LCOS) clearance can be reduced by as much as 3 times.^{7,14}
- See [above](#) for side effects common to all opioids

Special Considerations:

- Consider lower starting doses in patients with renal failure, with reduced hepatic function, with LCOS and in neonates.
- Consider rotation to fentanyl if severe bronchospasm or itch.
- If using a bolus of morphine prior to a painful procedure (e.g. ETT suctioning) remember that time to peak effect is 20 minutes after administration so allow sufficient time between the bolus dose and the procedure.

Fentanyl

See [Fentanyl medication monograph](#) for further details

Fentanyl is a synthetic opioid and is 70-100 times more potent as an analgesic than morphine but with minimal sedative activity.^{7,32} It has the advantage of a quicker onset of action and shorter duration of action following a single bolus dose as well as being more haemodynamically stable.¹⁴ It causes minimal histamine release and blunts systemic and pulmonary haemodynamic response to pain.^{1,7} These properties make it the preferred opioid in patients in PCC post cardiac surgery; with renal impairment; and with asthma.

Indications

- Haemodynamically unstable patients
- Patients post cardiac surgery
- Severe bronchoconstriction
- Significant renal impairment
- Significant side effects from morphine

Pharmacokinetics:

- Single bolus dose - short onset and offset due to rapid redistribution:
 - Time to onset: 30 seconds.¹⁴
 - Peak effect: 5-8 minutes.³³
 - Duration of effect 30-45 mins.¹⁴
 - Half-life 2 hours.³⁴
- Continuous infusion:
 - Context sensitive half-life of 21 hours following prolonged infusion.¹⁴
- Metabolised in liver to inactive metabolites which are then excreted in urine.¹⁴
- Clearance markedly reduced by reduced hepatic blood flow e.g. LCOS, hypothermia.¹⁴
- Clearance highest in children aged 6 months to 6 years of age.³³
- Unlike morphine it is not dependent on hepatic maturation to produce analgesia as it does not have active metabolites.¹⁴

Dosing for intubated patients:

- Loading dose: 2-3microg/kg IV (max 150microg)
- Continuous infusion dose: 0-4microg/kg/hr. Higher doses may be required, e.g. in opioid tolerant patients or patients on ECMO.
- Bolus dose for pain: 1-2microg/kg IV (max 50-100microg)³²

Adverse side effects:

- Tolerance develops more rapidly compared to morphine and there is a higher incidence of dependence and withdrawal.¹⁴
- Prolonged duration of action following cessation of continuous infusion compared with morphine due to longer elimination half-life.¹⁴
- Cardiovascular. Whilst haemodynamically stable at low doses it can cause bradycardia and decreased cardiac output in children with depressed cardiac function or hypovolaemia.¹⁴
- Significant accumulation can occur in patients with reduced hepatic perfusion, e.g. LCOS or increased intra-abdominal pressure.⁷
- Glottic and chest wall rigidity can occur with rapid bolus of higher doses (>3microg/kg).¹¹ May require management with muscle relaxation if severe.
- ECMO: Increased sequestration (adsorption onto circuit) may occur in patients on ECMO with up to 70% of dose lost in circuit requiring higher doses initially.¹³ Clearance may also be reduced due to reduced hepatic blood flow and withdrawal can occur with ECMO discontinuation due to restoration of hepatic blood flow and increased hepatic metabolism.¹¹
- see [above](#) for side effects common to all opioids

Hydromorphone

Hydromorphone is the third opioid that is used as an infusion in PCC. It is not generally used as a first line infusion but is useful for opioid rotation to minimise tolerance. Hydromorphone is 5 times more potent than morphine.^{13,35}

Pharmacokinetics:

- Bolus IV dose:
 - Time to onset: 5 mins.³⁶
 - Peak effect: 10-20 mins.³⁶
 - Duration of action 4-6 hours.⁷
- Elimination half-life 3 - 4 hours.⁷
- Metabolised by liver to inactive metabolites³⁵

Dosing for intubated patients

- Loading dose 10-20microg/kg IV (max 1mg)
- Infusion dose: 0-16microg/kg/hr
- Bolus dose for pain: 4-8microg/kg IV

Special Considerations

- May be preferred over morphine in patients with renal failure due to significantly less active metabolites.¹³
- Despite less histamine release, has been shown to have a similar rate of adverse effects including pruritus and nausea.³⁵
- Wide inter-individual pharmacokinetic variability, should be titrated to effect.⁷

Opioid rotation guidance

Rotation between opioids aims to reduce total opioid exposure in order to minimise development of tolerance and to minimise other adverse effects such as respiratory depression, feeding intolerance, constipation and ileus, withdrawal, hyperalgesia and immune system depression.^{8,37} Generally when rotating between opioids there is some novel effect hence a 25-50% reduction of total converted dose calculated can be used.²⁷ Opioid rotation should be considered after 5-10 days of continuous infusion. It may be considered sooner with fentanyl due to more rapid development of tolerance.

The following dose calculations for opioid rotation are used in PCC:

Table 1: Opioid Rotations in PCC

Current agent (microg/kg/hr)	New agent (microg/kg/hr)	Dose calculation
Morphine	Fentanyl	Divide by 20 then reduce by 25-50% for incomplete cross tolerance
Morphine	Hydromorphone	Divide by 5 then reduce by 25-50% for incomplete cross tolerance

Alpha 2 Agonists

These agents include dexmedetomidine and clonidine and have sedative, anxiolytic and weak analgesic effects but do not provide amnesia.^{38,39} They decrease the incidence of delirium and are used to manage the symptoms of delirium. They may have a neuroprotective effect. Minimal impact on respiratory drive or airway reflexes. Produce a sedative state from which a patient can be easily roused.

Dexmedetomidine

See [Dexmedetomidine medication monograph](#) for further details

Dexmedetomidine is the first line sedative in PCC and should be maximised (when tolerated) prior to the use of benzodiazepines. It has been shown to be safe in long term use in PICU and to prevent escalation of opioids and benzodiazepines when added to a sedation regimen.⁴⁰⁻⁴² Dexmedetomidine generally needs a separate vascular line and

by this may limit its use in some patients. In these patients intermittent enteral or IV clonidine or midazolam infusion can be used.

Indications

- First line sedative infusion in PCC for intubated patients
- Conscious sedation to allow patients to tolerate non-invasive ventilation (NIV)
- Prophylaxis or treatment of withdrawal or delirium
- Facilitation of extubation

Mechanism of action

- Acts on central alpha-2 adrenergic receptors to produce sedation and anxiolysis.¹³
- Highly selective CNS alpha-2-adrenergic receptor agonist with an alpha-2: alpha-1 specificity of 1600:1 (8 times more selective than clonidine).⁴³
- Analgesic effects due to stimulation of alpha-2 receptors in dorsal horn of spinal cord and acts centrally to potentiate opioid effects giving it significant coanalgesic properties.^{43,44}

Pharmacokinetics:

- IV bolus
 - Onset 30 minutes.¹³
 - Half-life of 2-2.65 hours.¹³
- Infusion:
 - Prolonged context sensitive half-life of 4hr after 8 hr infusion.³⁴
- Hepatic metabolism to inactive and non-toxic metabolites. Minimal renal contribution to elimination rate.⁴³
- Clearance in neonates is 40% of adult clearance, reaching 85% adult clearance rates by 1 year.¹⁴
- Clearance reduced by 25% on average in postoperative cardiac patients.¹⁴

Effects:

- Opiate and benzodiazepine sparing agent.^{38, 45-46}
- Respiratory system:
 - Minimal effect on respiratory drive or airway reflexes⁴³
- Cardiovascular:
 - May be useful in prevention and treatment of perioperative tachyarrhythmia e.g. junctional ectopic tachycardia (JET) ⁴⁷⁻⁴⁹

- Decreases myocardial demand for oxygen.⁴³
- Reduction in heart rate may lead to improved diastolic filling in patients with diastolic dysfunction, e.g. post Tetralogy of Fallot repair.⁴³
- CNS:
 - Produce a sedative state similar to natural sleep, allows for awakening for neurological assessment. May reduce development of delirium.⁴³
 - No effect on intracranial pressure (ICP) or seizure threshold.^{13, 46}
 - May reduced shivering in patients requiring cooling.⁴⁶
 - Preclinical evidence suggests that it may have a role in neuroprotection^{49,50}
- Other:
 - Less effect on gastric motility compared with opioids.¹³
 - Increase urine output and renal blood flow.⁴³

Adverse side effects

- Hypotension and bradycardia through effect on peripheral alpha-2 receptors. Resolves with dose reduction or fluid bolus. Not associated with CPR or permanent organ damage^{38, 51,52}
- Rebound hypertension and tachycardia on sudden cessation.^{40,44} Not linked to withdrawal episodes.
- Withdrawal syndrome on sudden cessation or rapid tapering.
 - Related to total dose and incidence around 30% in children who have received a dose equivalent to an infusion for greater than 5 days at 1microg/kg/hr.⁴⁴
- Symptoms include anxiety, agitation, decreased sleep, fever, loose stools, vomiting, tremors and increased secretions.^{40,44} There are no validated tools for the detection of dexmedetomidine withdrawal.

Dosing

- Used only as infusion in PCC without bolus dose. The maximum infusion rate is dependent on age:³⁸
 - 0-12 months: 1microg/kg/hr
 - 1-5 years: 1.4microg/kg/hr
 - 6-10 years: 1.3 microg/kg/hr
 - >10 years: 1microg/kg/hr

Special Considerations:

- Dosing adjustments needed in patients with hepatic impairment but not renal impairment¹³

Clonidine

See [Clonidine medication monograph](#) for further details

Clonidine is used for sedation, analgesia and control of symptoms of withdrawal symptoms from opioids or benzodiazepines (see [Withdrawal Syndrome Management guideline](#)). It is also used for conscious sedation, e.g. patients requiring non-invasive ventilation. It has not been shown to be linked to neurocognitive impairment.

Pharmacokinetics:

- Oral administration:
 - Onset 30-60 mins¹³
 - Time to peak effect 60-90 mins post oral administration.¹⁴
- Half-life 6 hours⁵³
- Alpha-2:Alpha1 receptor specificity of 200:1.⁴³
- Metabolised by liver to inactive metabolites and 50% excreted in urine/faeces as unchanged drug.⁵³
- Clearance in neonates is 1/3 adult rate due to immature clearance pathways and reaches 82% adult rate by 1 year of age.¹⁴

Dosing

- Enterally: 2-5 microg/kg 6-8 hourly (max 150microg).
- IV: 1-2microg/kg 6 hourly (max 50microg)

Key points

- Reduce dose in severe hepatic or renal impairment

Benzodiazepines

Benzodiazepines have dose dependent sedative, hypnotic, anxiolytic, amnestic, anticonvulsant and skeletal muscle relaxant properties. Importantly they have no analgesic effect.⁷ Benzodiazepines bind to the postsynaptic GABA receptors inhibiting the central nervous system. The side effects for all benzodiazepines include dose dependent respiratory depression (particularly when given in combination with opioids), hypotension, tolerance and withdrawal.

Midazolam

See [Midazolam medication monograph](#) for further details

Midazolam is used as a second line sedative infusion used in PCC after dexmedetomidine. It has well established efficacy as a sedative agent but has an increased risk of tolerance, withdrawal, delirium and possibly long term neurocognitive effects compared with dexmedetomidine. It produces antegrade amnesia without retrograde amnesia. Reduce cerebral metabolism and blood flow.⁷

Pharmacokinetics

- Bolus IV dose:
 - Onset 1-5 minutes³³
 - Peak effect 3-5 minutes^{13,34}
 - Duration of action 20-30 minutes^{13,34}
- Continuous infusion:
 - Elimination half-life 5.5 hours⁵⁴
 - Takes 20 hours to reach steady state if administered without a loading dose.⁵⁴
- Hepatic metabolism to active metabolites which are then renally eliminated.⁵⁴
- Maturity of hepatic metabolism doesn't reach adult levels until 3-12 months of postnatal age.⁵⁴
- Reduced clearance in critically ill children and in patients with renal or hepatic impairment.⁵⁴

Dosing for intubated patients

- Loading dose: 50-100microg/kg (max 5mg)⁴
 - Caution in patients with haemodynamic instability or post cardiac surgery, use a smaller dose and repeat if well tolerated.
- Infusion: 0-250microg/kg/hr (max 10mg/hr).⁴
- Bolus dose: 50microg/kg (max 2.5mg).⁴

Adverse effects

- Respiratory:
 - Produces a dose-related respiratory depression. This begins within 1-4 minutes of IV bolus and lasts for at least 30 minutes. Potentiated by concomitant opioids or other sedatives.⁷
- Cardiovascular:
 - Hypotension. Due to decreased systemic vascular resistance and decreased central sympathetic. Significant hypotension may occur in children with cyanotic congenital heart disease, hypovolaemia, with large bolus doses or when administered with opioids. Caution with loading doses in these patients.¹³
 - Decreased cardiac output: use of continuous infusions has been shown to result in reduction of cardiac output by >20% in post cardiac surgical patients.⁷

- Neurocognitive:
 - Strongly associated with the development of delirium.⁵⁵⁻⁵⁷
 - Associated with development of delusional memories and post-traumatic stress disorder.⁵⁸
 - Interferes with normal sleep architecture.⁷
 - Shown to cause apoptotic neurodegeneration in animal models.^{59,60}
 - Associated with long term negative neurodevelopmental outcome in children who have undergone cardiac surgery (cumulative dose-dependent).⁵
- Tolerance and withdrawal develop following prolonged use. A weaning plan should be commenced on any patient receiving 5 or more days of IV midazolam (see [Withdrawal Syndrome Management guideline](#)).⁶¹

Special Considerations

- Dose reduction may be required in hepatic and renal failure, haemodynamically unstable patients and patients with multi-organ dysfunction
- Careful use of bolus doses in patients with haemodynamic instability, hypotension, post-cardiac surgery.
- All patients on midazolam infusion for 5 days or longer should be weaned as per the [Withdrawal Syndrome Management Guideline](#)

Diazepam

See [Diazepam medication monograph](#) for further details

Has sedative, anxiolytic, amnestic and anticonvulsant properties and is longer acting compared to midazolam. Generally used in PCC to allow patients to wean from longer term midazolam infusions and manage withdrawal (see [Withdrawal Syndrome Management guideline](#))

Pharmacokinetics

- Onset 30-90 minutes post oral dose.⁷
- Variable half-life of 15-95 hours depending on age of child, with longer half-life in neonates and infants.¹³
- Undergoes hepatic metabolism to active metabolites with a long half-life.¹
- Metabolites are renally excreted.¹³

Adverse effects

- Dose dependent respiratory depression
- Mild reduction in systemic vascular resistance
- Prolonged use can lead to tolerance and withdrawal

Special Considerations

- Dose adjustment in hepatic and/or renal impairment

Anaesthetic agents**Propofol**

See [Propofol medication monograph](#) for further details.

Propofol provides sedation and hypnosis without analgesia. It also has amnestic, anxiolytic and anticonvulsant properties. Due to its short half-life it is a useful sedative agent pre-extubation to facilitate endotracheal tube tolerance while longer acting sedatives and analgesics wear off. Propofol is also useful for procedural sedation.

Indications

- Induction of anaesthesia
- Maintenance of sedation. Generally used peri-extubation and ideally should not be used for longer than 48 hours.
- Procedural sedation
- Anticonvulsant

Mechanism of action

- Centrally acting GABA agonist – binds to GABA_A subunit potentiating GABA-mediated synaptic inhibition within the central nervous system.⁶²

Pharmacokinetics

- Bolus dose
 - Rapid onset within 40 seconds due to its large volume of distribution with rapid redistribution to tissues.¹⁴
 - Peak effect in 1-5 minutes with a short duration of action of 2-8 minutes.¹
- Infusion
 - Prolonged half-life of propofol following continuous infusion with a context sensitive half-life of up to 40 minutes following an 8 hour infusion.³⁴
- Large volume of distribution accounts for the short duration of action of bolus doses.
- Increased clearance in children compared with adults.³⁴
- Metabolised by the liver to inactive metabolites that are then largely excreted by the kidneys.⁷

Dose

- Induction of anaesthesia: 1-3mg/kg. Titrate to effect.⁶²

- *NB: Avoid use in patients with haemodynamic instability or hypovolaemia*
- Infusion: maximum 4mg/kg/hr for 48 hours total dose including bolus doses.³⁴
- Bolus for inadequate sedation: 0.5-1mg/kg.^{13,34}
- Procedural sedation: 0.5-2mg/kg/hr.^{62,63} Start low and titrate up as required.

Effects

- Central nervous system: significant cerebral vasoconstriction with associated fall in cerebral blood flow, metabolic demand for oxygen, ICP and cerebral perfusion pressure. Anticonvulsant.^{62,63}

Adverse effects

- Respiratory: apnoea, loss of airway reflexes. Laryngospasm rare.⁶³
- Cardiovascular:
 - Decreased systemic vascular resistance and vasodilation, direct myocardial suppression, bradycardia. e.g. an induction dose of 2-2.5mg/kg propofol produces a 25-40% drop in systolic blood pressure.⁶³
 - Patients with hypovolaemia or cardiac dysfunction are more susceptible to significant hypotension with propofol.
 - Care should be taken in patients with hypovolaemia, shock, cardiac, renal or liver dysfunction.
- CNS: Animal studies suggest dose dependent neurotoxicity.⁶³
- Other: can also cause abnormal myoclonic movements that can be confused with seizures.⁷
- Hyperlipidaemia, especially in patients on parenteral nutrition, due to high lipid load (10% lipid emulsion) with long term use.
- Formulated in lipid emulsion which is a fertile media for bacterial proliferation leading to a risk of bacterial growth and sepsis.⁶³
- May exacerbate pancreatitis.

Propofol related infusion syndrome (PRIS)

- The major concern with use of propofol infusion in PICU is the risk of propofol related infusion syndrome, particularly if used at doses higher than 5mg/kg/hr (including bolus doses) or for duration greater than 48 hours but may occur at lower doses or shorter duration.
- This is an irreversible condition with a high mortality of up to 64%.⁶³
- The exact mechanism is unclear but related to depression of mitochondrial function leading to an imbalance between energy demand and utilisation.⁷

- Risk factors include dose >5mg/kg/hr (including bolus doses), duration of infusion >48 hours, younger age, critical illness, use of concomitant catecholamine infusions and/or steroids and inadequate carbohydrate load.^{34,64}
- It is characterised by an unexplained metabolic acidosis, ECG changes or rhabdomyolysis which can occur with or without acute kidney injury, hyperkalaemia, hyperlipidaemia, cardiac failure, fever, elevated liver enzymes or raised lactate.⁶⁴

Special Considerations

- No analgesic effect
- Maximum total dose (including bolus doses) should not exceed 4mg/kg/hr for maximum 48 hours due to risk of propofol related infusion syndrome.
- Use with caution in patients with hypovolaemia, shock or cardiac dysfunction due to risk of significant hypotension and decreased cardiac output.
- In those patients who do require propofol for longer than 48 hours then consider daily monitoring of pH, lactate and creatinine kinase.
- Consider PRIS in any patient on propofol with unexplained metabolic acidosis, new ECG changes or rhabdomyolysis.
- No need for dose adjustment for hepatic or renal impairment.
- Consider the lipid load (10% lipid emulsion) of infusion. Patients on parenteral nutrition should have triglyceride levels checked and may need reduction in rate of lipid infusions.
- Risk of bacterial infection: Meticulous aseptic technique; change line and syringe at least every 12 hours to minimise this.
- Avoid propofol in patients with underlying mitochondrial disorders.

Ketamine

See [Ketamine medication monograph](#) for further details

Ketamine has hypnotic, analgesic, sedative and antegrade amnestic properties. It has the benefits of having minimal effect on the respiratory drive and airway reflexes and being haemodynamically stable.¹⁴ It has a role as an adjunct analgesic and has been shown to be opioid sparing.¹³ It is also a bronchodilator making it a useful adjunct in patients with asthma.¹⁴

Indications

- Procedural sedation
- Induction of anaesthesia
- Analgesia adjunct– continuous infusion without bolus

Mechanism of action

- N-methyl-D-aspartate (NMDA) receptor antagonist resulting in potent analgesia, sedation, amnesia.
- It acts on cortex and limbic system to produce a dissociative sedation characterised by a profound analgesia and amnesia but not necessarily loss of consciousness.¹³
- Stimulates the release of endogenous catecholamines.

Pharmacokinetics

- Bolus dose:
 - IV: Onset of anaesthesia in 30-60 seconds^{7,13,14} Duration 5-10 minutes with recovery phase 1-2 hours^{7,14,65}
 - IM: Onset of anaesthetic effect in 3-4 minutes and duration of 12-25 minutes. Recovery period of 3-4 hours.⁶⁵
 - Enteral: Sedative effect in 10-15 minutes.⁶⁵
- Infusion:
 - Short context sensitive half-life of 40 minutes after 8 hour infusion.³⁴
- Hepatic metabolism initially to active then inactive metabolites. Inactive metabolites excreted in urine.

Dosing

- Procedural sedation:
 - IV: 1-2mg/kg. May repeat doses of 0.5-1mg/kg up to 5-15 minutely if required.⁶⁵
 - IM: 3-4mg/kg. May repeat doses of 2mg/kg after 10 minutes if required.⁶⁶
 - Enteral: 3-10mg given 30 minutes before the procedure.
- Induction of anaesthesia:
 - IV: 1-2mg/kg.⁶⁵
 - IM: 5-10mg/kg⁶⁵
- Analgesic adjunct:
 - IV infusion of 0-6microg/kg/min

Effects

- Adjunct analgesic with efficacy for neuropathic pain. Role as opioid sparing agent.¹³
- Respiratory: Minimal effect on respiratory drive and airway reflexes, bronchodilation.¹³

- Cardiovascular: Haemodynamically stable. Direct negative inotrope but this is usually offset by increase in sympathetic outflow. No effect on pulmonary artery pressures or pulmonary vascular resistance when normocarbida and normoxia maintained.³⁴
- Central nervous system: raises cerebral blood flow and cerebral oxygen demand. Does not appear to raise ICP.⁶⁵
- No clinical reports of withdrawal but tolerance has been reported.⁷

Adverse side effects

- Emergence delirium and hallucinations. Incidence is lower in children.¹³ Incidence may be minimised be reduced by concurrent administration of benzodiazepines or alpha-2 agonists in older children.⁶⁵
- Increase upper airway secretions.^{1,13} Can be managed with an anticholinergic agent, e.g. glycopyrrolate (glycopyrronium).
- Cardiovascular.
 - Can cause hypertension through increased sympathetic outflow.
 - Can cause hypotension and bronchoconstriction in patients who are catecholamine deplete, e.g. those with impaired myocardial contractility or severe shock.³⁴
- Laryngospasm
- Other: myoclonic movements⁷

Special Considerations

- Need for dose adjustment in severe hepatic impairment but not renal impairment.⁶⁸
- May cause hypotension and bradycardia in patients who are catecholamine depleted.

Sedative Adjuncts

Chloral hydrate

Chloral hydrate has sedative and hypnotic but no analgesic properties. It has minimal effect on respiratory drive in therapeutic single doses.

Indications

- Sedative adjunct for intubated patients
- Second line agent in children requiring conscious sedation after clonidine e.g. infants with bronchiolitis to improve tolerance of non-invasive ventilation (clonidine preferred due to lower risk of neurocognitive effects).
- Useful for procedural sedation for non-painful procedures (e.g. imaging, echocardiogram)

Mechanism of action

- Mechanism of action is unclear but acts similarly to barbiturates.⁶⁹

Pharmacokinetics

- Onset 10-15mins.⁶⁹
- Peak effect in 30-60 minutes⁶⁹
- Recovery time an additional 1-2 hours⁶⁹
- Rapid hepatic metabolism to trichloroethanol (TCE) which is the active drug.⁷⁰
 - Half-life of TCE is 8-12 hours but this is increased to 1-2 days in neonates.^{69,70}
- Further metabolised to trichloroacetate (TCA) which has some activity
- TCE and TCA are then excreted largely in the urine.⁷¹

Dosing

- Sedation adjunct
 - Enteral: 8-25mg/kg (max 500mg) 6-8 hourly PRN.⁷²
 - Titrate to effect based on patient response up to 50mg/kg/dose 6 hourly (max 500mg/dose).⁷²

Procedural sedation

- Single dose of 50mg/kg (max 1,000mg) 30 minutes prior to procedure.⁷²
- Best results in children < 3 years^{71,72}

Adverse effects

- Generally well tolerated at lower doses. At higher doses or with toxicity it can cause respiratory and cardiovascular effects^{70,72}
- Respiratory: Reduced respiratory drive, apnoea and airway obstruction may occur with high doses or with accumulation.⁷² Higher risk of respiratory compromise in patients with underlying obstructive sleep apnea.⁷²
- Cardiovascular: May cause hypotension. With toxicity or overdose myocardial depression and malignant cardiac arrhythmias can occur.⁷⁰
- Gastrointestinal: Nausea and vomiting, gastric irritation.^{69,70} Can result in haemorrhagic gastritis, gastric necrosis, enteritis and esophagitis with strictures in overdose.⁷⁰
- Neurological: Abnormal gait, ataxia, confusion, delirium, hallucinations can occur.⁷² Residual motor impairment and lethargy may last up to 24 hours post a single dose.⁷⁰ Possible long term neurocognitive effects related to cumulative dose.⁵

- Tolerance / withdrawal: Risk of withdrawal with sudden cessation. This can be similar to alcohol withdrawal with symptoms including delirium tremens and hallucinations.^{69, 71} Can be managed with slow weaning of doses or with clonidine.⁷³
- Hyperactivity / paradoxical excitement can occur in 6% of children.^{69,70}
- May cause hyperbilirubinaemia in neonates with prolonged use.⁷²
- Concern about possible carcinogenic and genotoxic effect in animals but not shown in human studies.⁷⁴

Special Considerations

- Avoid or reduce dosage in patients with marked hepatic or renal impairment.
- Avoid long term, regular use due to risk of accumulation of metabolite and toxicity.
- Monitor for conjugated hyperbilirubinaemia in newborns
- Taper dose gradually to minimise risk of withdrawal in patients who have been receiving regular chloral hydrate or use clonidine to manage withdrawal symptoms.^{71, 73}
- Increased risk of adverse events and treatment failures in children older than 4 years of age or with neurodevelopmental disorders.⁷⁵
- Best given with feeds to minimise gastric irritation.⁷⁰
- When used for procedural sedation patients must have cardiorespiratory monitoring.

Promethazine

A phenothiazine derivative which is a long-acting antihistamine with sedative, hypnotic and antiemetic effects.⁷⁶⁻⁷⁸ It also has mild anticholinergic and antiseritonergeric effects.⁷⁸

Pharmacokinetics

- Well absorbed into the GI system.⁷⁸
- Onset within 15-60 minutes of oral administration.⁷⁷
- Peak effect at 2-3 hours.⁷⁷
- Duration of sedative effects 2-8 hours.⁷⁷
- Extensive hepatic metabolism. Metabolites excreted in urine and bile. Elimination half-lives of 5-14 hours reported.⁷⁸

Dosing

- Enteral: 0.25-0.5mg/kg up to 6 hourly (max 25mg).^{76,77}

Adverse effects

- Respiratory depression (central and obstructive apnoea, reduced arousal), particularly in children aged <2 years old.⁷⁸
- Paradoxical reactions with agitation and hallucinations can occur.⁷⁸
- Other: seizures, extrapyramidal reactions, dystonic reactions, neuroleptic malignant syndrome, delirium, blood dyscrasias, hypotension and arrhythmias.^{76,77}

Special Considerations

- Avoid use in self-ventilating children <2 years old
- Caution with use in self-ventilating patients with obstructive sleep apnoea, altered conscious level or receiving other sedating medications
- Avoid in patients with severe hepatic impairment.⁷⁶

Phenobarbital (Phenobarbitone)

See [Phenobarbital medication monograph](#) for further details

Phenobarbital is a long acting barbiturate which has sedative, hypnotic and anticonvulsant effects.⁷⁹

Uses

- Sedation adjunct
- Refractory raised intracranial pressure
- Status epilepticus

Mechanism of action

- It produces sedation through inhibition of GABA sensitive chloride channels to produce global depression of the central nervous system

Pharmacokinetics

- Well absorbed from gastrointestinal tract (bioavailability 50-90%, lowest in neonates and infants).⁷⁹
- IV bolus:
 - Onset of action within 5 minutes.^{80,81}
 - Peak effect 30 minutes.^{80,81}
- Oral dose:
 - Onset of action 20-60 minutes.⁸⁰
 - Peak serum concentrations around 3 hours post dose.⁸¹

- Metabolised by liver to inactive metabolite. 25% excreted unchanged in urine and remainder excreted as inactive metabolites.⁸⁰
- Elimination half-life of around 3 days but up to 5 days in neonates.⁷⁹

Doses for sedation of intubated patients

- Initial IV loading dose: 5-10mg/kg (max 200mg). Can repeat up to 60mg/kg in 24 hours (max 1000mg).⁷⁹
- Regular dosing, IV/Enteral: up to 6mg/kg/day in 2-3 divided doses (maximum 100mg/day)⁷⁹⁻⁸².

Effects

- Decreased cerebral metabolic rate, reduction in cerebral blood flow and intracranial pressure.⁶⁸
- Potent anticonvulsant.⁶⁸

Adverse side effects

- Cardiovascular: Direct myocardial depressant and decreases systemic vascular resistance. Can cause severe hypotension resistant to inotropes at toxic doses.⁸⁰
- Respiratory depression: dose dependent.⁸¹
- Neurological: dose dependent sedation progressing to coma at toxic doses.⁸³
- Other: hypersensitivity reactions, Stevens-Johnson syndrome, withdrawal.⁸¹

Special Considerations

- Dose adjustment in severe hepatic and renal impairment.⁷⁹
- Vesicant medication when given IV. Avoid small veins and monitor injection site.
- In small doses can heighten the perception of pain and causes excitement rather than sedation ("antalgnesia").⁷
- Trough levels should be performed for patients who have received >40mg/kg IV phenobarbital. Significant toxicity associated with trough levels >80mg/L^{81, 83}
- CYP450 inducer – may accelerate metabolism of other medications.

Neuromuscular Blockade

Neuromuscular blocking agents (NMBA) are used in patients about to be intubated or who are already intubated. They do not have any sedative or analgesic properties and patients should be deeply sedated with an SBS of -2 to -3 prior to being muscle relaxed. A patient who has received a NMBA will no longer trigger the ventilator and should be changed from a spontaneous mode to a mandatory mode prior to receiving the dose of NMBA. Prolonged use of NMBA is associated with critical illness polyneuropathy and myopathy, particularly if used in combination with steroids.⁹ To reduce the risk of this it is important to use the minimum dose that is required.

Indications^{84,85}

- Intubation
- Severe pulmonary hypertension
- Refractory raised intracranial pressure
- Severe Acute Respiratory Distress Syndrome (ARDS)
- Severe haemodynamic instability (in particular low cardiac output state and junctional ectopic tachycardia)
- To control severe shivering during targeted temperature management
- To maintain tenuous cannula position in patients on ECMO
- Facilitation of procedures

Dosing

- Rocuronium and vecuronium are the most commonly used NMBA in PCC.
- Rocuronium is the most commonly used agent for intubation in PCC
 - Intubation: dose: 1-1.2mg/kg. Use the higher dose if rapid sequence intubation (RSI) required.
 - see [Rocuronium monograph](#) for further details
- Vecuronium is the most commonly used agent for intubated patients
 - may be used for intermittent paralysis at a dose of 0.1mg/kg PRN
 - see [Vecuronium monograph](#) for further details

Assessment of pain and sedation

- Patients require deep sedation (equivalent to SBS -3 or -2) and adequate analgesia while paralysed.
- The usual pain and sedation assessment tools are not validated for paralysed patients.
- Pupil response is not affected by neuromuscular blockade so dilated pupils can be an indicator of inadequate sedation.
- Significant physiological response to routine care is another indicator of inadequate analgesia or sedation. If a patients vital signs (HR, RR) increase by >20% with stimulation then it should be assumed that they have inadequate sedation or analgesia and they should be managed appropriately.⁴

NMBA continuous infusion

- An infusion of NMBA is rarely used in PCC and should always be discussed with the consultant prior to being commenced.

- If an infusion is considered clinically indicated then vecuronium is the preferred agent – this is renally cleared so it is important to remember that clearance may be prolonged in patients with renal failure.
- All patients managed with a continuous infusion in PCC should have:
 - Regular peripheral nerve stimulation (at least once per shift) aiming for a train of 4 result of 1 or higher (see [Peripheral Nerve Stimulator guideline](#)).⁸⁴
 - Consideration of a daily NMBA holiday depending on patient clinical condition.⁸⁴
 - Daily review of the ongoing requirement for NMBA infusion.⁹
- A structured physiotherapy program should be considered for any patient receiving a prolonged infusion of NMBA.⁸¹⁴

Nursing care of the patient receiving NMBA

- Regular eye (4 hourly as minimum) care with lubrication and eyelid closure due to risk of corneal drying and potentially permanent corneal damage.⁸³
- 4 hourly mouth care
- Continuous monitoring of HR, blood pressure (invasive BP or 5-15 minutely non-invasive readings), core temperature, SpO₂ and ETCO₂.
- Indwelling catheter insertion. NMBA may directly affect the external urethral sphincter (skeletal muscle) worsening the urinary retention from opioids.
- Patients administered neuromuscular blockade are not able to cough or swallow; important to note when suctioning the oropharynx, nasopharynx and endotracheal tube.
- Train of 4 (TOF) should be performed each shift with NMBA infusion stopped if TOF=0 and not recommenced until TOF ≥1
- Pressure injuries: Paralysed patients are at high risk of pressure injuries. A pressure risk assessment should be completed daily and a prevention and management plan should be implemented.
- The use of neuromuscular blockade should be explained to the family.

Procedural Sedation

Procedural sedation exists across a continuum from a state of wakefulness to anxiolysis to moderate sedation (“conscious sedation”) to deep sedation to general anaesthesia. Individual variations in drug response are not always predictable, thus important to prepare for the possibility that the patient may pass into a greater depth of sedation than planned. Common painful procedures in PCC include insertion of vascular access lines; insertion and removal of chest drains; suctioning; removal of wires, lumbar punctures.

Simple Procedures

- A bolus of the opioid infusion that the patient is currently receiving is often used for a simple uncomfortable procedure such as suctioning.

- Fentanyl is the preferred opioid for procedural analgesia because of its rapid onset and short duration.
- Consider use of topical local anaesthetic where appropriate.
- Distraction techniques should always be considered in awake children.

Moderately-severe painful procedures

- For example, insertion and removal of chest drains, insertion of central vascular access devices or arterial lines.

Fasting

- Non-intubated patients should ideally be fasted as per fasting guidelines for intubation where time permits
 - Infants <12 months: 1 hour clear fluids, 3 hours breast milk, 4 hours breast milk substitute, 6 hours solid food
 - Children >12 months: 1 hour clear fluids, 6 hours other fluids / food

Personnel:

- The minimum personnel should include the proceduralist in addition to an airway doctor and nurse. Consultant and nurse co-ordinator should be aware.

Monitoring:

- ECG, SpO2 monitoring, 2 minutely non-invasive BP monitoring or via arterial line, RR. Monitoring should continue until the patient has returned to baseline conscious state.

Equipment:

- Airway trolley should be prepared as there is a risk of deterioration requiring intubation with all procedural sedation.
- Intubation checklist should be completed prior to procedure.

Pharmacological options for procedural sedation for awake patients include:

- Ketamine: 1-2mg/kg IV initially and titrated to effect.⁶⁶
- Propofol: 0.5-2mg/kg IV initially and titrated to effect.^{62,63}
- Midazolam:
 - 0 - 5 years: IV initially 0.05 - 0.1mg/kg, titrated to effect to max 0.6mg/kg.⁸⁶
 - 6 -18 years: IV initially 0.025 - 0.05mg/kg, titrated to effect to max 4mg.⁸⁹
- Dexmedetomidine: IV 0.5-1microg/kg/hr. NB: this takes around 30 minutes to achieve steady state without a loading dose.
- Local anaesthetic should be used for all painful procedures (see below) even if the patient is intubated and ventilated.

Useful tips:

- Propofol and midazolam have no analgesic effects and should be used with local anaesthetic +/- opioid or ketamine for painful procedures.
- Depending on vein size, injection of propofol into peripheral veins is often painful. Injection of lidocaine 1% 0.05mL/kg or 0.5mg/kg (max 25mg) with occlusion of the vein prior to injection of propofol has been shown to reduce pain and be superior to injection of lignocaine-propofol admixture.⁶⁴

Local Anaesthetic Agents

- Local anaesthetics are drugs that reversibly block conduction of neural impulses along central and peripheral nerve pathways
- Local anaesthetic should be used for all painful procedures as it will reduce the requirement for systemic analgesia and sedation .
- Topical local anaesthetic such as lidocaine 2.5% with prilocaine 2.5% cream (e.g. EMLA[®]) can be used for less painful procedures, e.g. peripheral cannulation or lumbar puncture.
- In PCC lidocaine 1% (i.e. 50mg/5mL) without adrenaline (epinephrine) is the preferred local anaesthetic
- lidocaine 1% with adrenaline 1:100,000 (10microgram/mL) may be preferred to reduce bleeding risk secondary to vasoconstriction but should never be injected into areas of end-artery perfusion e.g. the penis or digit
 - **NB: Maximum dose of lidocaine 1% without adrenaline is 0.5mL/kg (i.e. 5mg/kg) or with adrenaline 1:100,000 is 0.7mL/kg (7mg/kg) .⁷**

References and related external legislation, policies, and guidelines	
1.	Zuppa AF, Curley MAQ. Sedation Analgesia and Neuromuscular blockade in Pediatric Critical Care. <i>Pediatr Clin N Am.</i> 2017;64:1103-1116.
2.	Harris J, Ramelet A-S, va Dijk M et al. Clinical recommendations for pain, sedation, withdrawal and delirium assessment in critically ill infants and children: an ESPNIC position statement for healthcare professionals. <i>Int Care Med.</i> 2016; 42:972-986.
3.	Poh YN, Poh PF, Buang SNH, Lee JH. Sedation Guidelines, Protocols and Algorithms in PICUs: A Systematic Review. <i>Pediatr Crit Care Med.</i> 2014;
4.	Curley MAQ, Wypij D, Watson RS et al. Protocolized Sedation versus Usual Care in Pediatric Patients Mechanically Ventilated for Acute Respiratory Failure: A Randomised Clinical Trial. <i>JAMA.</i> 2015;313(4):379-389
5.	Garcia Guerra G, Robertson CM, Alton GY et al. Neurotoxicity of sedative and analgesia drugs in young infants with congenital heart disease: 4 year follow up. <i>Pediatr Anesth.</i> 2014;24(3)257-265

References and related external legislation, policies, and guidelines	
6.	Williams C, Bratton SL. More science for the art of dexmedetomidine administration in pediatric critical care. <i>Mierva Anesthesiologica</i> . 2019;85(2):118-120
7.	Kudchadkar SR, Easley B, Brady KM and Yaster M. Pain and Sedation Management. In: Shaffner D, Nichols DG, editors. <i>Rogers Textbook of Paediatric Intensive Care</i> . 5 th ed. Philadelphia: Wolters Kluwer; 2015
8.	Traube C, Silver G, Gerber G et al. Delirium and Mortality in Critically Ill children: epidemiology and Outcomes of Pediatric Delirium. <i>Crit Care Med</i> . 2017 May;45(5): 891-898
9.	Vincent J-L, Shehabi Y, Walsh T et al. Comfort and patient-centred care without excessive sedation: the eCASH concept. <i>Int Care Med</i> . 2016;42:962-971
10.	Playfor S, Jenkins I, Boyles C et al. Consensus guidelines for sustained neuromuscular blockade in critically ill children. <i>Pediatric Anesthesia</i> . 2007;17:881-887.
11.	Vet NJ, Brusse JM, de Hoog M. et al. Inflammation and Organ Failure Severely Affect Midazolam Clearance in Critically Ill Children. <i>Am J Resp Crit Care Med</i> . 2016;194(1):58-66.
12.	Sucrose [Internet]. Australian Medicines Handbook Children's Dosing Companion. Jul 2021. Accessed online 2021 Nov 23.
13.	Johnson PN, Miller JL, Hagemann TM. Sedation and Analgesia in Critically Ill Children. <i>AACN Advanced Critical Care</i> . 2012;23(4):415-434
14.	Lucas SS, Nasr VG, Ng AJ et al. Pediatric Cardiac Intensive Care Society 2014 Consensus Statement. <i>PCCM</i> . 2016;17(3):S4-S15
15.	Acetaminophen (paracetamol): Pediatric drug information [Internet]. UpToDate; 2021 [cited 2021 Apr 28]. Available from: UpToDate
16.	Achuff B-J, Moffett BS, Acosta S et al. Hypotensive Response to IV Acetaminophen in Pediatric Cardiac Patients. <i>PCCM</i> . 2019.;20(6):527-533.
17.	Kokki H. Nonsteroidal Anti-Inflammatory Drugs for Postoperative Pain: A Focus on Children. <i>Pediatr Dugs</i> . 2003;5(2):103-123
18.	NSAIDS [Internet]. Australian Medicines Handbook. Jan 2021. Accessed online 2021 Apr 29
19.	Barbagallo M, Sacerdote P. Ibuprofen in the treatment of children's inflammatory pain: a clinical and pharmacological overview. <i>Minerva Pediatrica</i> . 2019;71(1):82-99
20.	Celecoxib: Pediatric drug information [Internet]. UpToDate; 2021 [cited 2021 Apr 28]. Available from: UpToDate
21.	Ibuprofen [Internet]. Australian Medicines Handbook Children's Dosing Companion. Jan 2021. Accessed online 2021 Apr 28.

References and related external legislation, policies, and guidelines
22. Celecoxib [Internet]. Australian Medicines Handbook Children's Dosing Companion. Jan 2021. Accessed online 2021 Apr 28.
23. De Martino M, Chiarugi A, Boner A et al. Working Towards an Appropriate Use of Ibuprofen in Children: An Evidence-Based Appraisal. <i>Drugs</i> . 2017;77:1295-1311
24. Hullet B, Salman S, O'Halloran SJ et al. Development of a Population Pharmacokinetic Model for Parecoxib and Its Active Metabolite Valdecoxib after Parenteral Parecoxib Administration in Children. <i>Anesthesiology</i> . 2012 May;116(5):1124-1133
25. Egunsola O, Wylie CE, Chitty KM, Buckley NA. Systematic Review of the Efficacy and Safety of Gabapentin and Pregabalin for Pain in Children and Adolescents. <i>Anesthesia Analgesia</i> . 2018;128(4):811-819
26. Gabapentin: Pediatric drug information [Internet]. UpToDate; 2021 [cited 2021 Apr 28]. Available from: UpToDate
27. Gabapentin [Internet]. Australian Medicines Handbook Children's Dosing Companion. Jan 2021. Accessed online 2021 Apr 28.
28. Parikh JM, Amolenda P, Rutledge J et al. An update on the safety of prescribing opioids in paediatrics. <i>Expert Opin Drug Saf</i> . 2019 Feb;18(2):127-143.
29. Cravero JP, Agarwal R, Berde R et al. The Society for Pediatric Anesthesia recommendations for the use of opioids in children during the perioperative period. <i>Pediatric Anesthesia</i> . 2019 Mar;29:547-571.
30. Anand KJS, Clark AE, Willson DF et al. Opioid Analgesia in Mechanically Ventilated Children: results from the multicentre MOTIF study. <i>Pediatr Crit Care Med</i> . 2013 Jan;14(1):27-36
31. Bouwmeester NJ, Anderson BJ, Tibboel D, Holford NHG. Developmental pharmacokinetics of morphine and its metabolites in neonates, infants and young children. <i>BJA</i> . 2004;92(2):208-17.
32. Thigpen JC, Odle BL, Hariforoosh S. Opioids: A Review of Pharmacokinetics and Pharmacodynamics in Neonates, Infants and Children. <i>Eur J Drug Met Pharm</i> . 2019;44:591-609
33. Ziesenitz VC, Vaughns JD, Koch G et al. Pharmacokinetics of fentanyl and its derivatives in children: a comprehensive review. <i>Clin Pharmacokinet</i> . 2018 Feb;57(2):125-149.
34. Nasr VG, DiNardo JA. Sedation and Analgesia in Pediatric Cardiac Critical Care. <i>PCCM</i> . 2016;17(8):S225-S231
35. Spénard S, Gélinais C, Trottier ED et al. Morphine or hydromorphone: which should be preferred? A systematic review. <i>Arch Dis Child</i> . 2020;0:1-8
36. Hydromorphone: Pediatric drug information [Internet]. Up to date; 2021 [cited 2021 Apr 28]. Available from: UpToDate.

References and related external legislation, policies, and guidelines
37. Sanavia E, Mencia S, Lafever SN et al. Sedative and Analgesic Drug Rotation Protocol in Critically Ill Children with Prolonged Sedation: Evaluation of Implementation and Efficacy to Reduce Withdrawal Syndrome. PCCM. 2019;20(12):1111-1117.
38. Erickson SJ, Millar J, Anderson BJ et al. Dexmedetomidine Sedation in Mechanically Ventilated Critically Ill Children: A Pilot Randomised Controlled Trial. PCCM. 2020;21(9):e731-e739
39. Walker T, Kudchadkar SP. Pain and Sedation Management: 2018 Update for the Rogers' Textbook of Pediatric Intensive Care. PCCM. 2019;20(1):54-61
40. Whalen LD, Di Gennaro JL, Irby GA, Yanay O, Zimmerman JJ. Long term dexmedetomidine use and safety profile among critically ill children and neonates. Pediatr Crit Care Med. 2014;15(8):706-714
41. Czaja AS, Zimmerman JJ. The use of dexmedetomidine in critically ill children and neonates. Pediatr Crit Care Med. 2009;10(3):381-86
42. Grant MJ, Schneider JB, Asaro LA et al. Dexmedetomidine Use in Critically Ill Children with Acute Respiratory Failure. Pediatr Crit Care Emd. 2016;17(12):1131-41.
43. Kiski D, Malec E, Schmidt C. Use of dexmedetomidine in pediatric cardiac anaesthesia. Curr Opin Anesthesiol. 2019;32:334-342
44. Haenecour AS, Seto W, Urbain CM et al. Prolonged Dexmedetomidine Infusion and Drug Withdrawal in Critically Ill Children. J Pediatr Pharmacol Ther. 2017;22(6):453-460.
45. Gupta P, Whiteside W, Sabati A et al. Safety and Efficacy of prolonged dexmedetomidine use in critically ill children with heart disease. PCCM 2012;13:660-666
46. Bejian S, Valasek C, Mlgro JJ et al. Prolonged use of dexmedetomidine in the CICU. Cardiol Young 2009;19:98-104
47. El Amrousy DM, Elshmaa NS, El-Kashlan M et al. Efficacy of prophylactic dexmedetomidine in preventing postoperative junctional ectopic tachycardia after paediatric cardiac surgery. J AM Heart Assoc. 2017;6(3):e004780
48. Li X, Zhang C Dai D et al. Efficacy of dexmedetomidine in prevention of junctional ectopic tachycardia and acute kidney injury after pediatric cardiac surgery: a meta-analysis. Congenit Heart Dis. 2018;13:799-807
49. Liu Y, Bian W, Liu P et al. Dexmedetomidine improves the outcomes in pediatric cardiac surgery: a meta-analysis of randomized controlled trials. Interact Cardiovasc Thorac Surg. 2018;26:852-858
50. Lin EP, Lee J-R L, Lee CS et al. Do anesthetics harm the developing human brain? An integrative analysis of animal and human studies. Neurotoxicology and Teratology. 2017;60:117-128
51. Daverio M, Sperotto F, Zanetto L et al. Dexmedetomidine for Prolonged Sedation in the PICU: A Systematic Review and Meta-Analysis. PCCM. 2020;21(7):e467-e474

References and related external legislation, policies, and guidelines
52. Sperotto F, Mondardini MC, Dell'Oste C et al. Efficacy and Safety of Dexmedetomidine for Prolonged Sedation in the PICU: A Prospective Multicentre Study (PROSDEX). <i>Pediatr Crit Care Med</i> . 2020. 21(7):625-636
53. Clonidine: Pediatric drug information [Internet]. UpToDate; 2021 [cited 2021 Apr 28]. Available from: UpToDate
54. de Wildt SN, de Hoog M, Vinks AA et al. Population pharmacokinetics and metabolism of midazolam in pediatric intensive care patients. 2003;31(7):1952-1958
55. Smith HAB, Gangopadhyay M, Goben C et al. Delirium and Benzodiazepines Associated with Prolonged ICU Stay in Critically Ill Infants and Young Children. <i>Crit Care Med</i> . 2017;45(9):1427-1435.
56. Mody K, Kaur S, Mauer EA et al. Benzodiazepines and development of delirium in critically ill children: Estimating the causal effect. <i>Crit Care Med</i> . 2018;46:1486-1491
57. Traube C, Silber G, Reeder RW et al. Delirium in critically ill children: An international point prevalence study. <i>Crit Care Med</i> 2017;45:584-590
58. Colville G, Kerry S, Pierce C. Children's Factual and Delusional Memories of Intensive Care. <i>Am J Respir Crit Care Med</i> . 2007;177:976-982
59. Young C, Jevtovic-Todorovic V, Qin YQ et al. Potential of ketamine and midazolam, individually or in combination, to induce apoptotic neurodegeneration in the infant mouse brain. <i>Br J Pharmacol</i> 2005;146:189-197
60. Jevtovic-Todorovic V, Hartman RE, Izumi Y et al. Early exposure to common anesthetic agents causes widespread neurodegeneration in the developing rat brain and persistent learning deficits. <i>J Neurosci</i> . 2003;23:876-882.
61. Best KM, Wypif D, Asaro LA et al. Patient, Process and System Predictors of Iatrogenic Withdrawal Syndrome in Critically Ill Children. <i>Crit Care Med</i> . 2017;45(1):e7-e15
62. Propofol: Pediatric drug information [Internet]. UpToDate; 2021 [cited 2021 Nov 23]. Available from: UpToDate
63. Chidambaran V, Costandi A, D'Mello A. Propofol: A review of its role in pediatric anesthesia and sedation. <i>CNS Drugs</i> . 2015;29(7):543-563.
64. Hemphil S, McMenamin L, Bellamy LC, Hopkins PM. Propofol infusion syndrome; a structured literature review and analysis of published case reports. <i>Br J Anaest</i> . 2019;122:448-59
65. Ketamine [Internet]. Australian Medicines Handbook Children's Dosing Companion. Jul 2021. Accessed online 2021 Nov 23. Ketamine: Pediatric drug information [Internet]. UpToDate; 2021 [cited 2021 Aug 24]. Available from: UpToDate
66. Ketamine [Internet]. Australian Medicines Handbook Children's Dosing Companion. Jul 2021. Accessed online 2021 Nov 23

References and related external legislation, policies, and guidelines
67. Zeiler FA, Teitelbaum J, West M et al. The ketamine effect on ICP in traumatic brain injury. <i>Neurocrit Care</i> . 2014;21(1):163-73.
68. Tobais JD. Sedation and analgesia in Pediatric intensive care units: a guide to drug selection and use. <i>Pediatr Drugs</i> . 1999;1:19-26
69. Chloral hydrate: NeoFax/Paediatric monograph[Internet]. Micromedix. Accessed online 2021 Apr 28..
70. Pershad J, Palmisano P, Nichols M. Chloral hydrate: The good and the bad. <i>Pediatr Emerg Care</i> . 1999;15(6):432-435
71. Chloral hydrate. In: MIMS Online [Internet]. St Leonards (New South Wales): MIMS Australia; 2020 [cited 2021 March 23]. Available from: https://mimsonline.com.au
72. Chloral hydrate: Paediatric drug information[Internet]. UpToDate. Accessed online. 2021 Apr 28. Available from: UpToDate
73. Da Silva PS, Passos RMA, Waisberg DR. Withdrawal treatment with clonidine after prolonged use of chloral hydrate in a pediatric intensive care patient.
74. Haselkorn T, Whittemore AS, Udaltsova N, Friedman GD. Short-Term Chloral Hydrate Administration and Cancer in Humans. <i>Drug Safety</i> . 2006;29(1):67-7
75. Cozzi G, Norbedo S, Barbi E. Intranasal Dexmedetomidine for Procedural Sedation in Children, a Suitable Alternative to Chloral Hydrate. <i>Pediatr Drugs</i> . 2017;19:107-111
76. Promethazine: Pediatric drug information [Internet]. UpToDate; 2021 [cited 2021 Apr 28]. Available from: UpToDate
77. Promethazine: Paediatric drug information. [Internet]. Clinical pharmacology. Accessed 2021 Apr 28. Available from: Clinical Pharmacology.
78. Promethazine: Product disclosure information [Internet]. MIMS; 2021 [cited 2021 Apr 28]. Available from: MIMS
79. Phenobarbital: Pediatric drug information [Internet]. UpToDate; 2021 [cited 2021 Apr 28]. Available from: UpToDate
80. Phenobarbitone: Product disclosure information [Internet]. MIMS; 2021 [cited 2021 Apr 28]. Available from: MIMS
81. Phenobarbitone: Pediatric drug information. [Internet]. Clinical pharmacology. Accessed 2021 Apr 28. Available from: Clinical Pharmacology.
82. Phenobarbitone: Paediatric drug information. [Internet]. Micromedix Accessed 2021 Apr 28. Available from: Micromedix.
83. Murray L, Little M, Pascu O, Hogget K. <i>Toxicology Handbook</i> . 3 rd edition. Published 2015.

References and related external legislation, policies, and guidelines	
84.	Renew JR, Ratzlaff R, Hernandez-Torres V et al. Neuromuscular blockade management in the critically ill. J Intensive Care. 2020;8:37
85.	Bernier ML, Romer L, Bembea M, Spectrum of Current Management of Pediatric Pulmonary Hypertensive Crisis. Crit Care Expl. 2019 Aug;1(8):e0037
86.	Krauss B, Green SM. Procedural sedation and analgesia in children. Lancet. 2006 Feb;367:766-780.
87.	Ramelet A-S, Rees NW, McDonald S et al. Clinical validation of the Multidimensional Assessment of Pain Scale. Pediatr Anesthes. 2007;17:1156-1165
88.	Malviya S, Voepel-Lewis T, Burke C, Merkel S, Tait A. The revised FLACC observational pain tool: improved reliability and validity for pain assessment in children with cognitive impairment. Pediatr Anaesth. 2006;16:258-265.
89.	Hicks CL, von Baeyer CL, Spafford PA et al. The Faces Pain Scale – Revised: toward a common metric in pediatric pain measurement. Pain. 2001 Aug;93(2):173-183
90.	von Baeyer CL, Spagrud LJ, McCormick JC et al. three new datasets supporting use of the Numerical Rating Scale (MRS-11) for children's self-reports of pain intensity. Pain. 2009;143(3):223-227
91.	Curley MAQ, Harris SK, Fraser KA, Johnson RA, Arnold JH. State Behavioral Scale (SBS) A Sedation Assessment Instrument for Infants and Young Children Supported on Mechanical Ventilation. Pediatr Crit Care Med. 2006;7(2):107-114

Related CAHS internal policies, procedures and guidelines	
	Withdrawal Syndrome Management guideline (Clinical Practice Manual)
	Paracetamol medication monograph
	Morphine medication monograph
	Fentanyl medication monograph
	Dexmedetomidine medication monograph
	Clonidine medication monograph
	Midazolam medication monograph
	Diazepam medication monograph
	Propofol medication monograph
	Ketamine medication monograph
	Phenobarbitone medication monograph

Related CAHS internal policies, procedures and guidelines
Rocuronium monograph
Vecuronium monograph
Peripheral nerve stimulator guideline
WAT-1 scoring MR 859.80
Pressure injury Prevention and Management Policy (CAHS Policy Manual)
Pressure Injury Prevention and Management Guideline (Clinical Practice Manual)

Useful resources (including related forms)
List and hyperlink the titles of useful resources, do not hyperlink MR forms

This document can be made available in alternative formats on request.

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Appendix 1: Algorithm for intubated Patients < 12 months

Target SBS as prescribed (usual -1 to 0), MAPS <4

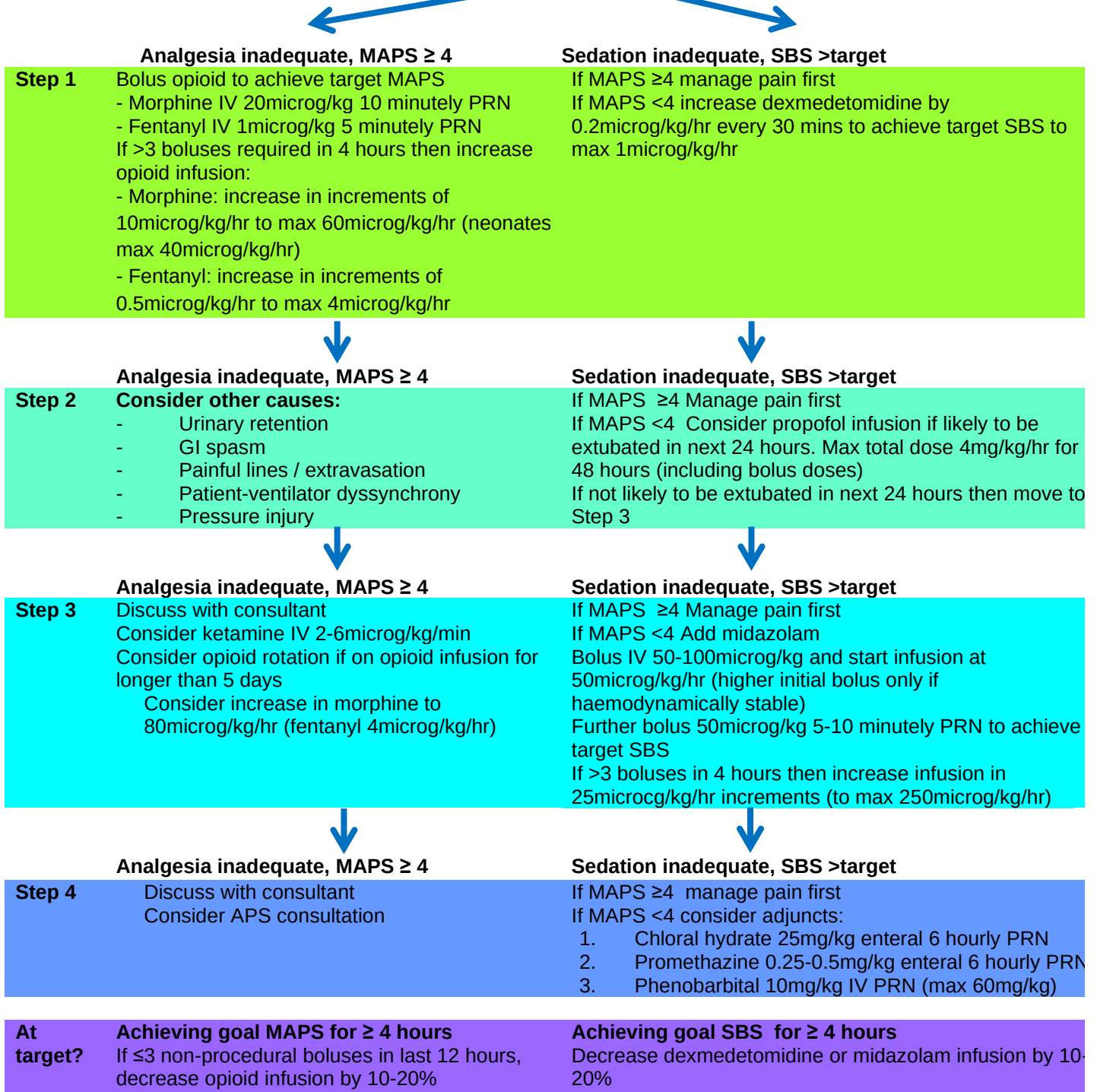
Initial sedation with: IV morphine 20microg/kg/hr (neonate 10microg/kg/hr) or IV fentanyl 2microg/kg/hr*

AND IV dexmedetomidine 0.5-1microg/kg/hr**

Give loading dose of morphine 50-100microg/kg or fentanyl 2-3microg/kg if starting opioid infusion in PCC

Assess SBS and MAPS every 4 hours or 30 mins post PRN dose

Optimise simple analgesia: paracetamol, NSAIDs, sucrose



* Fentanyl first line opioid in patients post cardiac surgery, with renal failure or asthma

** Midazolam to be used instead of dexmedetomidine as first line if contraindication to dexmedetomidine. Start as per plan from Step 2 inadequate sedation. If insufficient IV access makes dexmedetomidine difficult then consider regular enteral clonidine 2-5microg/kg 6 hourly as an alternative

Appendix 2: Algorithm for Intubated patients > 1-18 years, <50kg

Target SBS as prescribed (usual -1 to 0), MAPS<4

Initial sedation with: IV morphine 20microg/kg/hr or IV fentanyl 1microg/kg/hr*

AND IV dexmedetomidine 0.5-1microg/kg/hr**

Give loading dose of morphine 50-100microg/kg or fentanyl 2-3microg/kg/ if starting opioid infusion in PCC

Assess SBS and MAPS every 4 hours or 30 minutes post PRN dose

Optimise simple analgesia: regular paracetamol + NSAIDs (if appropriate)

Optimise non-pharmacological measures: Noise control, parental presence, verbal and non-verbal communication, frequent repositioning, day-night routine, music therapy



Analgesia inadequate, MAPS ≥ 4

Step 1

Bolus opioid to achieve target MAPS:

- morphine IV 20microg/kg 10 minutely PRN
- fentanyl IV 1microg/kg 5 minutely PRN

If >3 boluses required in 4 hours then increase opioid infusion:

- Morphine: increase in increments of 10microg/kg/hr (max 60microg/kg/hr)
- Fentanyl: increase in increments of 0.5microg/kg/hr (max 4microg/kg/hr)

Sedation inadequate, SBS >target

If MAPS ≥ 4 manage pain first

If MAPS <4 increase dexmedetomidine by 0.2microg/kg/hr every 30 mins to achieve target SBS to age appropriate maximum#:

- 1-5 years: 1.4microg/kg/hr
- 6-10 years: 1.3microg/kg/hr
- >10 years: 1microg/kg/hr



Analgesia inadequate, MAPS ≥ 4

Step 2

Consider other causes:

- Urinary retention
- GI spasm
- Painful lines / extravasation
- Patient-ventilator dyssynchrony
- Pressure injury



Sedation inadequate, SBS >target

If MAPS ≥ 4 : Manage pain first

If MAPS <4: Consider addition of propofol infusion if likely to be extubated in next 24 hours. . Maximum total dose 4mg/kg/hr for 48 hours (including bolus doses)

If not likely to be extubated in next 24 hours then move to Step 3



Analgesia inadequate, MAPS ≥ 4

Step 3

Discuss with consultant

Consider ketamine IV 2-6 microg/kg/min
Consider opioid rotation if on opioid infusion for longer than 5 days
Consider gabapentin if neuropathic pain
Consider increase in morphine to max 80microg/kg/hr



Sedation inadequate, SBS >target

If MAPS ≥ 4 Manage pain first

If MAPS <4 Add midazolam

Bolus IV 50-100microg/kg and start infusion at 50microg/kg/hr (higher initial bolus only if haemodynamically stable)

Further bolus 50microg/kg 5-10 minutely PRN to achieve target SBS

If >3 boluses in 4 hours then increase infusion by 25 microg/kg/hr (to max 250 microg/kg/hr)



Analgesia inadequate, MAPS ≥ 4

Step 4

Discuss with consultant
Consider APS consult



Sedation inadequate, SBS >target

Consider adjuncts

1. Chloral hydrate 25mg/kg (max 500mg) enteral 6 hourly
2. Promethazine 0.25-0.5mg/kg 6 hourly enteral PRN
3. Phenobarbital 10mg/kg IV PRN (max 60mg/kg)

At target

Achieving goal MAPS for ≥ 4 hours

If ≤ 3 non-procedural boluses in last 12 hours, decrease opioid infusion by 10-20%

Achieving goal SBS for ≥ 4 hours

Decrease dexmedetomidine or midazolam infusion by 10-20%.

* Fentanyl first line opioid in patients post cardiac surgery, with renal failure or asthma

** Midazolam to be used instead of dexmedetomidine as first line if contraindication to dexmedetomidine. Start as per plan from Step 2 inadequate sedation. If insufficient IV access makes dexmedetomidine difficult then consider regular enteral clonidine 2-5microg/kg (max 150microg) 6 hourly as an alternative.

Appendix 3: Algorithm for intubated patients > 50kg

Target SBS as prescribed (usual -1 to 0), MAPS <4

Initial sedation with: IV morphine 1-2mg/hr* or IV fentanyl 50microg/hr*

AND IV dexmedetomidine 0.5-1microg/kg/hr**

Give loading dose of morphine 2.5-5mg or fentanyl 100-150microg if starting opioid infusion in PCC

Assess SBS hourly and MAPS every 4 hours or 30 mins post PRN dose

Optimise simple analgesia: paracetamol and NSAIDs (if appropriate)

Optimise non-pharmacologic measures: Noise control, parental presence, verbal and non-verbal communication, frequent repositioning, day-night routine, music therapy



Analgesia inadequate, MAPS ≥ 4

Step 1 Bolus opioid to achieve target MAPS:
 - morphine IV 1mg 10 minutely PRN
 - fentanyl IV 50microg 5 minutely PRN
 If >3 boluses required in 4 hours then increase opioid infusion:
 - Morphine: increase in increments of 0.5mg/hr (max 4mg/hr)
 - Fentanyl: increase in increments of 25microg/hr (max 200microg/hr)

Sedation inadequate, SBS >target, MAPS<4

If MAPS ≥4 manage pain first
 If MAPS <4 increase dexmedetomidine by 0.2microg/kg/hr every 30 mins to achieve target SBS to maximum 1microg/kg/hr



Analgesia inadequate, MAPS ≥ 4

Step 2 Consider other causes:
 - Urinary retention
 - GI spasm
 - Painful lines / extravasation
 - Patient-ventilator dyssynchrony
 - Pressure injury

Sedation inadequate, SBS >target, MAPS<4

Consider addition propofol infusion if likely to be extubated in next 24 hours. Maximum dose of 4mg/kg/hr (including boluses) for maximum of 48 hours. Otherwise move to Step 3



Analgesia inadequate, MAPS ≥ 4

Step 3 Consider ketamine IV 2-6microg/kg/min
 Consider gabapentin if neuropathic pain
 Consider tramadol
 Consider opioid rotation if on opioid infusion for longer than 5 days

Sedation inadequate, SBS >target, MAPS<4

If MAPS ≥4 Manage pain first
 If MAPS <4 Add midazolam.
 Bolus IV 2.5-5mg and start infusion at 2.5mg/hr Higher initial bolus only if haemodynamically stable.
 Further bolus 2.5mg 5-10 minutely PRN to achieve target SBS. If >3 boluses in 4 hours increase infusion by 1mg/hr (max 10mg/hr)



Analgesia inadequate, MAPS ≥ 4

Step 4 Discuss with consultant
 Increase opioid infusion incrementally as per step 1 until pain controlled
 Consider APS consultation

Sedation inadequate, SBS >target, MAPS<4

Consider adjuncts:
 1. Phenobarbital 100-200mg IV PRN (max 1g)
 2. Promethazine 12.5-25mg enteral 8 hourly PRN
 3. Chloral hydrate 250-500mg enteral 6 hourly PRN

At target? **Achieving goal MAPS for ≥ 8 hours**
 If ≤3 non-procedural boluses in last 8 hours, decrease opioid infusion by 10-20%

Achieving goal SBS for ≥ 8 hours
 If ≤3 non-procedural boluses of midazolam and achieving target SBS for last 8 hours then decrease dexmedetomidine or midazolam infusion by 10-20%.

* Fentanyl as first line opioid in patients post cardiac surgery, with renal failure or asthma

** Midazolam to be used instead of dexmedetomidine as first line if contraindication to dexmedetomidine. Start as per plan from Step 3 inadequate sedation. If insufficient IV access makes dexmedetomidine difficult then consider regular enteral clonidine 50-150microg 6 hourly as an alternative.

Appendix 4: MAPS (Multidimensional Assessment of Pain Scale)

The PICU MAPS (multidimensional assessment of pain scale) was designed especially for the critically ill infant and child. It has been validated in children aged 0-3 years.⁸⁷ A total score of 3 or less is considered optimal for analgesia.

Categories	0	1	2	Score
Vital Signs HR and/or BP	Within Baseline	HR >10bpm and/or Systolic/Diastolic BP >10mmHg above baseline	HR >10bpm below baseline AND Systolic/Diastolic BP >10mmHg above baseline	
Breathing pattern	No change	Development of increased work of breathing, rate or paradoxical breathing	Respiratory distress with silent or weak cry	
Facial expressions	Relaxed	Occasional grimace	Frequent grimace – may be associated with silent or weak cry	
Body movements	No movements or purposeful movements	Restless – jerky leg movements, twitching hands or feet, twisting or writhing	Rigid and/or limited body movements (incl. pulling up knees, spreading feet, clenched fist)	
State of arousal	Calm or asleep – awakens easily and is not distressed	Hyper-reactive – excessively responsive to non-painful stimuli (light, noise)	Lethargic – decreased physical alertness and activity, no eye contact	
Total score				

Appendix 5: FLACC - R (Revised – Faces, Legs, Activity, Cry, Consolability)

- Use for children from birth to eight years and older children not able to communicate or self-report including children with a developmental disability (i.e. nonverbal or preverbal children).⁸⁸
- Evaluates age-appropriate behavioural and physiological factors
- Observe the child for 1 minute, uncover legs if able; add the scores from each of the five measurement categories and document a total score of 0-10.
- *Italic text are additional descriptors validated in children with cognitive impairment. Individualised behaviours, described by carers, can be added to the score of 2 for those with developmental disability*

Categories	Score 0	Score 1	Score 2
FACE	No particular expression or smile	Occasional grimace / frown, withdrawn or disinterested; <i>appears sad or worried</i>	Consistent grimace or frown; frequent constant quivering chin, clenched jaw, <i>distressed looking face; expression of fright or panic</i> Individualised behaviours: _____
LEGS	Normal position or relaxed; <i>usual tone and motion to limbs</i>	Uneasy, restless or tense; <i>occasional tremors</i>	Kicking, or legs drawn up; <i>marked increase in spasticity, constant tremors or jerking.</i> Individualised behaviours: _____
ACTIVITY	Lying quietly, normal position, moves easily; <i>regular, rhythmic respirations</i>	Squirming, shifting back and forth, <i>tense or guarded movements; mildly agitated, (e.g. head back and forth, aggression; shallow, splinting respirations, intermittent sighs</i>	Arched, rigid or jerking, <i>severe agitation, head banging, shivering (not rigors), breath-holding, gasping or sharp intake of breath, severe splinting.</i> Individualised behaviours: _____
CRY	No cry / verbalisation	Moans or whimpers, occasional complaint; <i>occasional verbal outburst or grunt</i>	Crying steadily, screams or sobs, frequent complaints; <i>repeated outbursts, constant grunting.</i> Individualised behaviours: _____
CONSOLABILITY	Content and relaxed	Reassured by occasional touching, hugging or being talked to. Distractible.	Difficult to console or comfort; <i>pushing away caregiver, resisting care or comfort measures</i> Individualised behaviours: _____

Appendix 6: Pain Self-Report Assessment Tools

Faces Pain Scale – Revised (FPS-R)⁸⁹

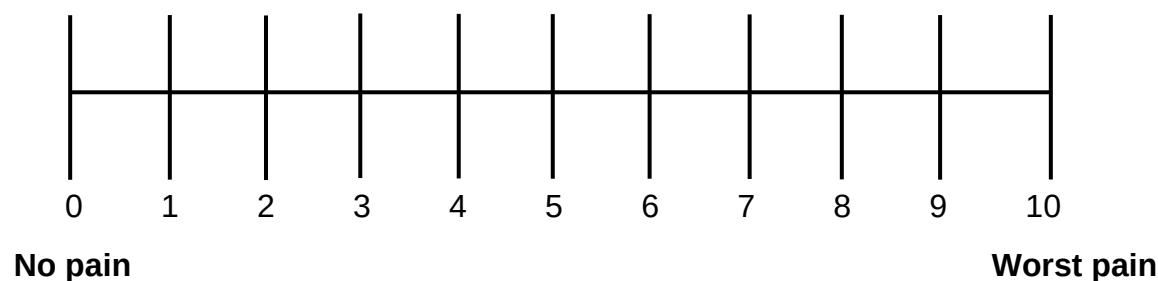
- Can be used from ages 4 and over
- Say “hurt” or pain” whichever seems right for a particular child
- Instructions: “These faces show how much something can hurt. This face [point to left-most face shows no pain. The faces show more and more pain [point to each from left to right] up to this one [point to right-most face] – it shows very much pain. Point to the face that shows how much you hurt [right now]”.



- Score the chosen face 0,2,4,6,8 or 10, counting left to right, so ‘0’ = ‘no pain’ and ‘10’ = ‘very much pain’. Do not use words like ‘happy’ or ‘sad’. This scale is intended to measure how children feel inside, not how their face looks

Numeric pain rating scale⁹⁰

- Use from age seven years and older
- Best used as a visual tool until 12 years and after this can be used as a verbal tool



Appendix 7: State Behavioural Scale (SBS)

- Validated in mechanically ventilated children aged 6 weeks to 6 years of age
- Score as patient's response to voice then gentle touch then noxious stimuli (planned endotracheal suctioning or < 5 seconds of nail bed pressure)

Score	Description	Definition
-3	Unresponsive	No spontaneous respiratory effort No cough or coughs only with suctioning No response to noxious stimuli Unable to pay attention to care provider Does not distress with any procedure (including noxious) Does not move
-2	Responsive to noxious stimuli	Spontaneous yet supported breathing Coughs with suctioning / repositioning Responds to noxious stimuli Unable to pay attention to care provider Will distress with a noxious procedure Does not move / occasional movement of extremities or shifting of position
-1	Responsive to gentle touch or voice	Spontaneous but ineffective nonsupported breaths Coughs with suctioning / repositioning Responds to touch / voice Able to pay attention but drifts off after stimulation Distresses with procedures Able to calm with comforting touch or voice when stimulus removed Occasional movement of extremities or shifting of position
0	Awake and able to calm	Spontaneous and effective breathing Coughs when repositioned / Occasional spontaneous cough Responds to voice / No external stimulus is required to elicit response Spontaneously pays attention to care provider Distresses with procedures Able to calm with comforting touch or voice when stimulus removed Occasional movement of extremities or shifting of position / increased movement (restless, squirming)
+1	Restless and difficult to calm	Spontaneous effective breathing / Having difficulty breathing with ventilator Occasional spontaneous cough Responds to voice / No external stimulus is required to elicit response Drifts off / Spontaneously pays attention to care provider Intermittently unsafe Does not consistently calm, despite 5 minute attempt / unable to console Increase movement (restless, squirming)
+2	Agitated	May have difficulty breathing with ventilator Coughing spontaneously No external stimulus required to elicit response Spontaneously pays attention to care provider Unsafe (biting ETT, pulling at lines, cannot be left alone) Unable to console Increased movement (restless, squirming or thrashing side to side, kicking legs)